

SYSTEMATIC EXPERIMENTAL DATA ON THE DRAG OF SHARP AND BLUNT CONES AT HYPERSONIC SPEEDS

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Results are presented for systematic tests of circular sharp and spherically blunted cones. Data are obtained on the drag coefficient C_x in

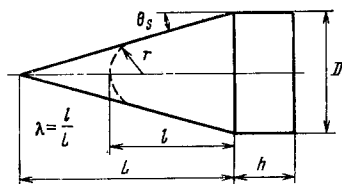


Fig. 1

the form of parametric grids with respect to the cone half-angle θ_s and the fineness ratio λ , for Mach numbers from 8 to 14. The Reynolds numbers were on the order of $(7-12) \cdot 10^6$, i. e., they corresponded to the flow regions in which the viscous interaction effect can be neglected. The results are compared with the theoretical calculations of [1, 2] and the experimental data from [3-7].

The tests were conducted by firing models into an opposing supersonic wind-tunnel stream. The testing technique was described briefly in [8]. The table shows the geometric parameters of the models tested and the notation is illustrated in Fig. 1. The models without the cylindrical shank ($h/D = 0$) were fired from a pallet while the cone models with a cylinder ($h/D = 0.533$) were fired from a gun barrel.

We note that the slope of the oblique shock on the sharp cones, with a half-angle which does not exceed 50° , corresponds to the theoretical values of [1]. The flow past the sharp cone with $\theta_s = 70^\circ$ for $M = 9.8$ and $R = 9.4 \cdot 10^6$ is accompanied by a detached shock wave (Fig. 2). The curvature and location of the shock in the vicinity of the stagnation point of the blunted cone is determined completely by the shape of the corresponding spherical segment. Figure 3 shows the flow pattern about the blunt cone with $\theta_s = 15^\circ$ and $\lambda = 0.8$ for $M = 9.5$ and $R = 5.5 \cdot 10^6$.

The drag coefficient C_x versus M for sharp and blunt cones with angle $\theta_s = 15^\circ$ is shown in Fig. 4; the points 1, 2, 3, and 4 correspond to the values $\lambda = 0.4, 0.6, 0.8$, and 1.0. With increase in M , the drag coefficient decreases, and, for $M \geq 8$, it remains practically constant.

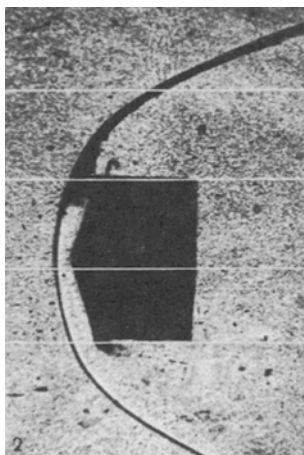


Fig. 2

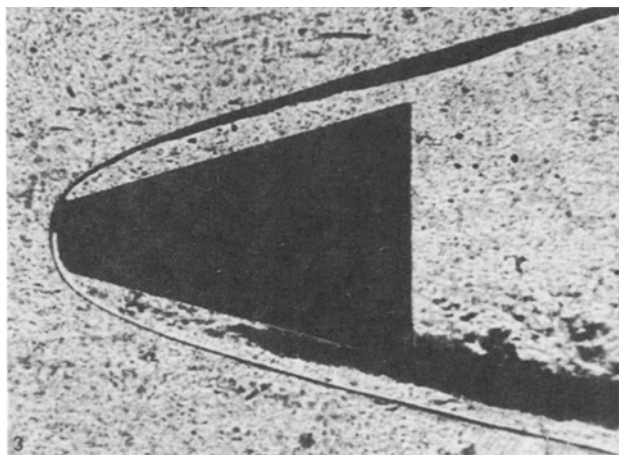


Fig. 3

Figure 5 shows the experimental curves of C_x for the cones as a function of the angle θ_s (for various values of the slenderness ratio λ in the range $M = 8-14$), when the drag coefficient is independent of M . These relations form the region of values of the drag coefficients for any possible sharp or blunt cones. Solid curves are drawn through the experimental values and the values obtained by interpolation of the experimental data. Dot-dashed curves represent the experimental values of the drag for the spherical segments into which the limiting blunt cones degenerate. The dashed curves correspond to the blunt cone drag coefficient C_{x+} obtained as the sum of the experimental values of the drag of the spherical nose segment and the conical surface.

The interference of the blunt nose with the conical surface,

$$\Delta C_x = \frac{C_x}{C_{x+}} - 1,$$

depends on the cone half-angle and slenderness ratio. The maximum value of the interference will be on the order of 10% for $\lambda = 0.7$, which we see from Fig. 6, where curves 1, 2, 3, and 4 correspond to the values $\theta_s = 15, 24, 35$, and 50° . Estimates show that the over-all fraction of the frictional- and base-pressure drag amounts to 3%, which lies within the limits of experimental accuracy ($\pm 5-7\%$). Hence, the good agreement of the experimental data with the theoretical values of the cone drag coefficient from [2] (Fig. 5—open triangles), obtained without account for frictional and base pressure.

In the range of moderate values of the cone half-angle (up to $40-50^\circ$) two parameters—the half-angle θ_s and the slenderness ratio λ —have a significant influence on the value of the drag coefficient. There is practically no effect of slenderness ratio on the drag coefficient for large values of the angle θ_s . In this case, the drag of the sharp cones coincides with that of the corresponding inscribed spherical segments.

Experimental curves of the drag coefficient for the models tested, as a function of the slenderness ratio λ , and, for various angles θ_s , are presented in Fig. 7. The limiting values of the model drag coefficient for the angle $\theta_s = 0$, shown by the dashed curve, were taken from Fig. 5, for $\theta_s = 0$.

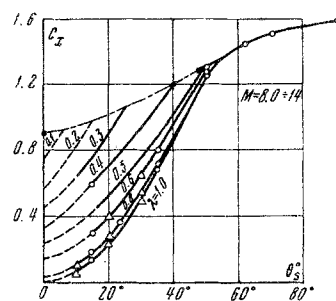


Fig. 4

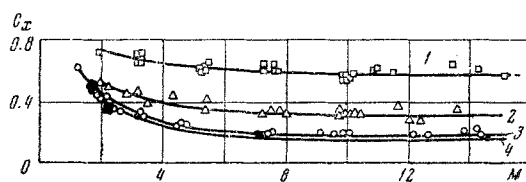


Fig. 5

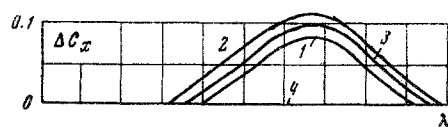


Fig. 6

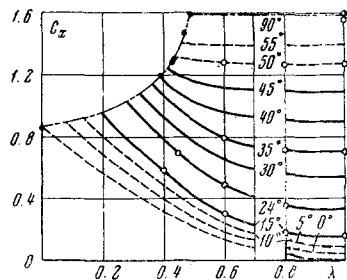


Fig. 7

θ°_s	λ	τ/D	h/D	θ°_s	λ	τ/D	h/D
15	1.0	0	0	35	1.0	0	0
15	0.8	0.131	0	35	0.8	0.19	0.533
15	0.6	0.261	0.533	35	0.6	0.38	0.533
15	0.4	0.391	0.533	50	1.0	0	0.533
24	0.8	0.153	0.533	50	0.8	0.275	0.533
24	0.6	0.306	0.533	50	0.6	0.549	0.533
24	0.45	0.422	0.533	70	1.0	0	0.533

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